**Project Report: Sales Forecasting & Demand Prediction**

**Project Name:** End-to-End Sales Forecasting with Machine Learning & Power BI

**Date:** 01-02-2026

**Tools Used:** Python (Pandas, Scikit-Learn, Prophet), Power BI

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## 1. Executive Summary

The objective of this project was to build a comprehensive end-to-end solution to predict future daily sales for a retail chain and visualize key business insights. Using historical sales data (2013–2017), we performed data cleaning, exploratory data analysis (EDA), and time-series modeling.

We trained and compared two distinct models: **Facebook Prophet** and **Random Forest Regressor**. The **Random Forest** model proved superior, achieving a **MAPE (Mean Absolute Percentage Error) of 6.67%**, significantly outperforming the baseline. The final deliverables include a robust Python pipeline for 30-day future forecasting and an interactive Power BI dashboard that allows management to track KPIs, identifying high-performing regions and underperforming product categories.

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## 2. Problem Statement & Objective

Retail businesses often struggle with inventory mismanagement due to inaccurate demand planning. Overstocking leads to waste, while understocking leads to lost revenue.

- **Goal:** To analyze historical sales data and forecast demand for the next 30 days.
- **Scope:** The project covers data cleaning, feature engineering, model selection, and the creation of a Business Intelligence (BI) dashboard for decision-making.

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## 3. Data Processing Pipeline

We utilized a multi-table dataset including train.csv (sales logs), stores.csv (metadata), and holidays_events.csv.

### 3.1 Data Cleaning & Preparation

- **Handling Missing Values:** Imputed missing values in the sales and onpromotion columns with 0 to maintain data integrity.
- **Date Conversion:** Converted the date column to DateTime objects to enable time-series operations.

- **Outlier Removal:** Identified and removed negative sales entries (returns/errors) to prevent model skew.
- **Merging:** Joined store metadata (City/State) with the main transactional data to enable regional analysis.

### 3.2 Feature Engineering

To enable the Random Forest model to "learn" time dependencies, we created the following features:

- **Lag Features:** sales_lag_7 (Sales 1 week ago) and sales_lag_30 (Sales 1 month ago).
- **Time Components:** Day of week, Month, Year, and "Is Weekend" flags.
- **Rolling Statistics:** 7-day rolling mean and standard deviation to capture short-term trends.

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## 4. Machine Learning Model Explanation

We tested two different approaches to find the most accurate forecasting method:

### Model A: Facebook Prophet

- **Type:** Additive Time-Series Model.
- **Strengths:** Excellent at handling seasonality (yearly/weekly) and holidays.
- **Result:** While it captured the general trend, it struggled with sharp daily fluctuations driven by promotions.

### Model B: Random Forest Regressor

- **Type:** Ensemble Decision Tree Algorithm.
- **Why it was chosen:** Unlike Prophet, Random Forest can ingest external features like "Promotion Status" and "Lagged Sales," allowing it to react dynamically to recent sales changes.
- **Result:** This model captured complex non-linear relationships better than the statistical model.

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## 5. Forecast Accuracy Metrics

We evaluated both models on a hold-out test set (last 166 days of data).

| Metric | Facebook Prophet | Random Forest (Winner) |
| --- | --- | --- |
| **MAE** (Mean Absolute Error) | 73,112 | **60,551** |
| **RMSE** (Root Mean Squared Error) | 109,034 | **85,936** |
| **MAPE** (Mean Absolute % Error) | 7.76% | **6.67%** |

**Conclusion:** The Random Forest model reduced the error rate by over **1%** compared to Prophet. A MAPE of **6.67%** indicates high reliability for inventory planning.

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## 6. Business Insights & Dashboard

The final Power BI dashboard consolidates technical forecasts into actionable business intelligence.

**Key Insights Generated:**

1. **Top Performing Categories:**
    - **"GROCERY I"** and **"BEVERAGES"** are the primary revenue drivers, accounting for the bulk of daily sales. These require high inventory priority.
2. **Underperforming Categories:**
    - Categories like **"BOOKS"**, **"BABY CARE"**, and **"MAGAZINES"** consistently show low volume. These are candidates for inventory reduction or discontinuation.
3. **Regional Performance:**
    - **Pichincha** is the dominant state for sales volume, followed by Guayas. Logistics hubs should be centered near these high-demand zones.
4. **Seasonality:**

+1

- Sales show a clear spike in **December** (Year-end holidays). The Year-Over-Year matrix confirms consistent growth in Q4 across all years (2013–2016).

**Dashboard Visuals:**

- **Timeline Forecast:** A continuous line chart showing 4 years of history + 30 days of future predictions.
- **Leaderboards:** Bar charts displaying the "Top 5" and "Bottom 5" product families.
- **Geospatial Analysis:** A Treemap/Map visualizing sales density across different states.

- **YoY Matrix:** A heatmap table comparing monthly performance across years.

*[Insert Screenshot of your Power BI Dashboard Here]*

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## 7. Recommendations for Management

Based on the data analysis and ML forecasts, we recommend:

1. **Inventory Optimization:** Increase stock levels for "Grocery I" and "Produce" by 10% in anticipation of the forecasted 30-day trend.
2. **Strategic Cuts:** Reduce shelf space for "Books" and "Hardware" in low-volume stores to save operational costs.
3. **Regional Marketing:** Focus promotional budgets on the **Pichincha** region, as it yields the highest ROI per transaction.
4. **Holiday Planning:** Prepare supply chains for a heavy surge in demand starting in late November, as indicated by the historical seasonality analysis.

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## 8. Conclusion

This project successfully demonstrated how machine learning can transform raw sales data into a strategic asset. By implementing a Random Forest regressor, we achieved a highly accurate forecast (93.3% accuracy), empowering the business to make data-driven decisions regarding inventory and sales strategy.